# FARDIN AHSAN SAKIB

Fairfax, VA 571-668-1676 fsakib@gmu.edu Website LinkedIn US Permanent Resident

**EDUCATION:**

**Ph.D. Candidate, Computer Science** — **August 2021–Present**
George Mason University — **Fairfax, VA**

**M.S. in Computer Science** — **August 2021–May 2024**
George Mason University — **Fairfax, VA**

**Bachelor's in Computer Science and Engineering** — **January 2017–March 2021**
Islamic University of Technology — **Dhaka, Bangladesh**

**EXPERIENCE:**

**AI Research Intern** — **June 2026–August 2026**
Comcast (Xfinity Assistant, Applied AI) — **Washington, DC**

- Designed and built cross-session memory for a multi-agent LLM assistant that adapts agent behavior for repeat customers, raising prior-context acknowledgment from 22% to 89%.
- Created an LLM-as-judge evaluation pipeline to measure memory quality and agent behavior, and used it to improve fact coverage from 45% to 80%.

**Agentic AI Research Intern** — **November 2025–January 2026**
Fujitsu Research of America — **Santa Clara, CA**

- Engineered the candidate-selection stage for test-time scaling of LLM coding agents on SWE-Bench Verified, picking the best of 8 agent-generated patches using cross-agent test results.
- Trained an LLM verifier (87% accuracy) and compared it with test-based selection, showing that executable test evidence ranks candidate patches more reliably.

**Applied Research Scientist Intern** — **June 2025–August 2025**
Amazon (AWS Support) — **Seattle, WA**

- Built a multi-step tool-calling benchmark from real AWS support queries that require chained, cross-service API calls, with a simulation server that replays recorded API responses for reproducible, cost-free agent evaluation.
- Benchmarked frontier LLM agents using step-level metrics and a validated LLM-as-judge, showing the best model resolved only ~20% of real support queries.

**Applied Research Intern** — **May 2024–August 2024**
Brillient Corporation — **Reston, VA**

- Implemented query distillation, chunking, and re-ranking for a retrieval-augmented generation (RAG) system, improving retrieval relevance by 22% (MAP) and cutting processing time by 35%.
- Fine-tuned domain-specific LLMs and deployed the system across medical, HR, and IRS domains, reaching 85% user satisfaction.

**Graduate Research Assistant** — **August 2021–Present**
George Mason University — **Fairfax, VA**

- Designed evaluation frameworks exposing reliability failures in LLMs, showing models rely on spurious features such as demographic cues, with up to 37-point error gaps across groups.
- Developed an inference-time correction method that reads the model's layer-by-layer prediction trajectory to decide which predictions to fix, correcting up to 79% of errors with low break rates across three model families and four datasets, with no retraining.
- Proposed a training-free defense for multi-agent LLM systems against prompt injection, reducing attack success by over 80% with under one point of accuracy loss and outperforming seven existing defenses.
- Led labs and discussions for 100+ students, providing individualized feedback on assignments and exams.

**TECHNICAL EXPERTISE:**

- **Research Areas:** Machine Learning, Deep Learning, Natural Language Processing (NLP), Large Language Models (LLMs), Foundation Models, Generative AI, Mechanistic Interpretability, Inference-time Steering, Post-training, Bias Mitigation, AI Safety, Model Alignment, Clinical NLP, Healthcare AI, Multi-Agent Systems, Prompt Injection, Adversarial Robustness
- **Methods:** Fine-tuning, LoRA, RAG, Verifier Models, Reward Modeling, RLHF, DPO, Model Evaluation, Benchmarking, Agentic AI, Coding Agents, Tool Use, LLM-as-Judge, Test-time Scaling
- **Frameworks & Tools:** PyTorch, Hugging Face, Transformers, LangChain, Multi-GPU Training, HPC Clusters
- **Programming:** Python, C++, SQL
- **Cloud & MLOps:** AWS (SageMaker, Lambda, API Gateway), Azure, Docker, Kubernetes
- **Data Engineering:** Vector Databases, Pandas, NumPy
- **Development Tools:** Git, API Development

**RECENT RESEARCH PUBLICATIONS:**

- **Sakib, Fardin Ahsan**, Ziwei Zhu, and Özlem Uzuner. "Trajectory-Guided Steering: Mitigating Spurious Feature Bias in Clinical Classification." *Under review*.
- **Sakib, Fardin Ahsan**, Ziwei Zhu, and Özlem Uzuner. "The Model Knows Better: Defending Medical Multi-Agent Systems Against Prompt Injection." *Under review*.
- **Sakib, Fardin Ahsan**, Ziwei Zhu, Karen Trister Grace, Meliha Yetişgen, and Özlem Uzuner. "Spurious Correlations and Beyond: Understanding and Mitigating Shortcut Learning in SDOH Extraction with Large Language Models." In Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (**ACL 2025**), pp. 1097–1106. (Link)
- Rahman, Md Mushfiqur, **Fardin Ahsan Sakib**, Fahim Faisal, and Antonios Anastasopoulos. "To token or not to token: A Comparative Study of Text Representations for Cross-Lingual Transfer." In Proceedings of the 3rd Workshop on Multi-lingual Representation Learning @ **EMNLP**, pp. 67–84. 2023. (Link)
- **Sakib, Fardin Ahsan**, AHM Rezaul Karim, Saadat Hasan Khan, and Md Mushfiqur Rahman. "Intent Detection and Slot Filling for Home Assistants: Dataset and Analysis for Bangla and Sylheti." In The First Workshop on Bangla Language Processing @ **EMNLP**, p. 48. 2023. (Link)